DBA-UxS

Uncrewed Systems: Threat Considerations for Critical Infrastructure

A Cross-Sector Foundational Instrument

Applicable to All Sectors Adopting the DBA Methodology

Version 1.0 — DRAFT

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This document applies to all sector implementations of the DBA methodology.

Revision History

Version	Date	Author	Description
1.0	April 2026	T. Roxey	Initial release. Generalizes the DBA-MA-UxS addendum (v1.0) into a cross-sector foundational instrument applicable to all critical infrastructure sectors adopting the DBA methodology. Removes nuclear-specific safety-function references from Sections 4–6 and 8; replaces them with sector-agnostic critical-infrastructure framing. Adds Section 10 establishing the sector implementation annex architecture. Nuclear-specific content migrated to DBA-MA-UxS-NA v1.0.
1.1	April 2026	Tim Roxey	DBA-O&G-UxS references updated throughout to reflect bifurcation into separate DBA-OIL (Petroleum Sector) and DBA-GAS (Natural Gas Sector) instruments. Sector list, forthcoming instrument entry, and references section updated accordingly.

1. Purpose and Scope

This document establishes the uncrewed systems (UxS) component of the DBA threat characterization, engineering design requirements, and monitoring and review framework for critical infrastructure sectors adopting the Design Basis Accident (DBA) methodology. It is a foundational cross-sector instrument applicable to all national and sector-specific implementations of the DBA series, including DBA-MA (Military Action against Nuclear Facilities), DBA-W (Water Sector), DBA-OIL (Petroleum Sector), DBA-GAS (Natural Gas Sector), DBA-DC (Data Centers), and future sector extensions.

This document generalizes and supersedes the nuclear-sector content of DBA-MA-UxS v1.0 with respect to its cross-sector elements. The nuclear-specific safety function references, facility vulnerability profiles, and parent document cross-references previously contained in DBA-MA-UxS v1.0 have been migrated to the DBA-MA-UxS-NA Nuclear Implementation Annex, which is a child document of this instrument. Readers implementing the DBA-MA series for nuclear facilities shall use this document in conjunction with DBA-MA-UxS-NA. Readers implementing other sector series shall use this document in conjunction with the applicable sector implementation annex.

The scope of this document is the full family of uncrewed systems that present credible threats to critical infrastructure safety functions under military action conditions. It does not address the governance of uncrewed military systems, the legality of their use under international law, or the national policy frameworks governing their development and deployment. Those questions are addressed in other instruments. This document addresses the threat to critical infrastructure facilities and the engineering and operational responses required to maintain safety and continuity of function under that threat.

The following sectors currently have or are developing sector implementation annexes that inherit from this document:

Nuclear (DBA-MA-UxS-NA v1.0): Applies the UxS threat framework to nuclear power plants and supporting nuclear infrastructure. Parent document: DBA-MA International Framework (L1, v1.4).

Water Sector (DBA-W-UxS, forthcoming): Applies the UxS threat framework to water treatment and distribution infrastructure. Parent document: DBA-W National Framework.

Data Centers (DBA-DC-UxS, forthcoming): Applies the UxS threat framework to hyperscale and critical data center infrastructure. Parent document: DBA-DC Design Basis Framework.

Petroleum Sector (DBA-OIL-UxS, forthcoming) and Natural Gas Sector (DBA-GAS-UxS, forthcoming): Applies the UxS threat framework to pipeline, terminal, and compressor station infrastructure. Parent documents: DBA-OIL Sector Framework (petroleum) and DBA-GAS Sector Framework (natural gas).

2. Taxonomy and Definitions

This document adopts the NATO and United States Department of Defense terminology for uncrewed systems as the authoritative reference. This alignment ensures that the threat characterizations and engineering requirements developed here are expressed in language that is immediately accessible to national defense establishments, alliance planning staffs, and the regulatory and technical organizations that must coordinate with them.

2.1 Uncrewed Systems (UxS)

Uncrewed Systems (UxS) is the umbrella term for the full family of uncrewed platforms across all operational domains. The “x” variable represents the domain of operation: aerial (A), surface maritime (S), and undersea (U). The term encompasses both remotely piloted platforms, in which a human operator maintains real-time control through a communications link, and autonomous platforms, in which onboard processing governs navigation, target identification, and engagement without continuous human direction. The distinction between remotely piloted and autonomous operation is relevant to countermeasure design: communications-link disruption is an effective countermeasure against remotely piloted systems, but is progressively less effective as autonomy increases.

2.2 Uncrewed Aerial System (UAS)

An Uncrewed Aerial System (UAS) is an aircraft and its associated elements — including ground control stations, data links, and support equipment — operated without a human pilot aboard. For DBA purposes, the relevant UAS categories are: one-way attack drones (loitering munitions), which are expendable platforms that carry an explosive payload to a designated target; first-person-view (FPV) platforms, which are lightweight, high-maneuverability platforms guided by an operator using a video feed, typically carrying a shaped-charge warhead and used for precision component-level targeting; reconnaissance platforms, which conduct persistent surveillance and pre-strike target identification; and autonomous AI-enabled systems, which integrate machine learning for target recognition, navigation, and jamming resistance, operating with reduced or no dependence on a communications link.

The conflict in Ukraine has established UAS as the primary instrument for military action against energy infrastructure at scale. The sustained drone campaign against Ukrainian generation, transmission, and distribution assets has demonstrated one-way attack volumes of dozens to hundreds of platforms per strike event, FPV platforms directed against specific infrastructure components and personnel, and AI-enabled autonomous systems achieving reported strike effectiveness rates of approximately eighty percent against hardened targets. This operational precedent is the foundation of the UAS threat characterization throughout this document.

2.3 Uncrewed Surface Vessel (USV)

An Uncrewed Surface Vessel (USV) is an autonomous or remotely piloted waterborne platform that operates on the water surface without a crew aboard. For DBA purposes, the relevant USV categories are: explosive-payload attack craft, which carry a substantial explosive charge — typically in the range of 75 to 450 kilograms — and are directed against waterfront infrastructure, vessel hulls, or intake structures; surveillance and reconnaissance USVs, which conduct persistent monitoring of waterfront approaches and facility waterside perimeters; and payload-delivery platforms, which can position equipment, sensors, or sub-systems in proximity to a facility without direct operator exposure.

The Ukraine conflict has provided the foundational operational precedent for USV use as a military instrument. Ukrainian naval drone operations against Black Sea Fleet assets beginning in 2022 demonstrated that commercially derived watercraft can be modified to carry militarily significant explosive payloads and directed against high-value targets with sufficient precision to cause severe structural damage. The Magura V5 and successor platforms demonstrate that USV capability is accessible to actors without traditional naval force-projection resources. The threat vector against critical infrastructure arises principally from facilities sited on navigable waterways, lakes, or coastal locations, where the same waterway that provides a process water source also provides an adversary with an unobstructed approach corridor.

2.4 Uncrewed Undersea Vehicle (UUV)

An Uncrewed Undersea Vehicle (UUV) is an autonomous or remotely piloted underwater platform that operates submerged without a crew. For DBA purposes, the relevant UUV categories are: explosive-payload attack platforms, which can approach submerged infrastructure — including water intake piping, discharge structures, and submerged cable runs — and deliver an explosive charge at close range or in contact; survey and reconnaissance platforms, which conduct subsurface mapping of facility waterside infrastructure in support of targeting; and payload-emplacement platforms, which can position sensors, mines, or sabotage devices on or near submerged infrastructure without direct attribution.

UUV capability at the state and near-state actor level is well established, with programs documented across multiple NATO and non-NATO states. At the sub-state and commercial-technology level, the miniaturization of autonomous undersea navigation and payload delivery represents a directionally significant capability trend. For facilities dependent on waterway-sourced cooling or process water, the intake structure is a safety-significant target accessible from the water and not addressed by above-surface physical protection measures.

2.5 Uncrewed Ground Vehicle (UGV)

Uncrewed Ground Vehicles (UGVs) are excluded from the scope of this document. UGV threats to critical infrastructure are addressed within the physical protection framework as an extension of the vehicle-borne improvised explosive device (VBIED) and ground-based intrusion threat categories. The relevant sector implementation documents govern this threat category. If operational developments produce UGV threats that present materially different characteristics from the existing physical protection framework, a separate addendum will be developed at that time.

3. Common Threat Characteristics Across the UxS Family

Despite operating in different physical domains, UAS, USV, and UUV share threat characteristics that distinguish them from other military instruments addressed in the DBA series. These common characteristics drive domain-agnostic requirements that should be reflected in the baseline engineering and operational posture of all facilities within the scope of the DBA framework.

3.1 Accessibility Without Traditional Force Projection

State and sub-state actors can employ all three UxS categories without the logistics infrastructure, personnel, or institutional structures required for conventional military operations. A UAS campaign against energy infrastructure requires launch vehicles, navigation systems, and payload integration, but not an air force in the traditional sense. A USV attack requires a watercraft, an explosive payload, and a communications system, but not a navy. A UUV operation requires autonomous navigation hardware and a payload, but not a submarine fleet. This accessibility characteristic means that the UxS threat is not bounded by assessments of the adversary’s conventional military capability. Facilities that are not plausible targets for conventional military attack may nonetheless be plausible targets for UxS operations by actors with limited resources and high motivation.

3.2 Operator Separation

All UxS categories create physical separation between the operator and the platform. This separation reduces the deterrent effect of physical protection measures directed at the operator: a defensive perimeter that would prevent a human attacker from approaching a facility does not prevent a UAS, USV, or UUV from being directed against it from a safe distance. Operator separation also reduces the observable indicators associated with an attack: a UAS strike can be initiated from a launch point kilometers away, a USV attack from a shore location beyond the facility perimeter, and a UUV operation from a vessel or launch point that may be geographically remote from the facility. The platform's approach — not the operator's presence — is the observable indicator that must trigger a protective response.

3.3 Persistent and Cumulative Capability

UxS operations can be sustained over time at costs that are low relative to the damage they inflict. The asymmetry between platform cost and consequence cost has been demonstrated operationally in the conflict in Ukraine. A Shahed-136 platform costs approximately $20,000 to $30,000 to produce; the replacement cost of the infrastructure it destroys is orders of magnitude higher. This cost asymmetry enables an adversary to sustain a UxS campaign for the duration of a military action, imposing cumulative damage through repeated strikes at the same location, targeting replacement equipment during installation, and degrading safety-significant systems progressively rather than in a single decisive event. The DBA methodology's requirement for engineering design that addresses sustained, deliberate degradation — rather than a single initiating event — is directly driven by the persistent, cumulative nature of UxS threats.

3.4 Reconnaissance-Strike Integration

All UxS categories can be used in both reconnaissance and strike roles, and the intelligence gathered by reconnaissance platforms directly supports targeting by strike platforms. A reconnaissance UAS can provide persistent aerial surveillance of a facility's layout, identifying the locations of safety-significant equipment, backup systems, consumable storage, and personnel shelters. A survey UUV can map the submerged infrastructure of a facility's process water intake. A USV conducting a slow-speed approach can document waterfront access routes, obstacles, and sensor positions. In each case, the reconnaissance information enables the adversary to identify specific vulnerabilities and direct strike platforms accordingly. This reconnaissance-strike cycle — in which the same technology that executes the attack also informs its targeting — means that engineering responses to UxS threats must address both the strike and the reconnaissance dimensions. Concealment and signature reduction are engineering requirements, not security preferences.

3.5 AI Integration and Countermeasure Degradation

The integration of artificial intelligence into UxS operations is advancing across all three domains, with materially different implications for countermeasure effectiveness. For UAS, AI enables autonomous target recognition, GPS-free navigation, resistance to radio-frequency jamming, and swarm coordination. For USV, AI enables autonomous navigation through complex waterway environments and target discrimination without real-time operator control. For UUV, AI enables autonomous underwater navigation without acoustic beacon support, making detection and tracking more difficult. As AI integration advances, countermeasures that rely on disrupting the operator control link — jamming, signal interference, geofencing — become progressively less effective. Engineering responses that do not depend on the effectiveness of active electronic countermeasures — physical hardening, passive kinetic barriers, concealment, and structural protection — offer greater durability than approaches that do.

4. Uncrewed Aerial Systems (UAS)

4.1 Threat Characterization

The UAS threat to critical infrastructure encompasses the full range of uncrewed aerial platforms documented in Section 2.2. For threat-class definition and hardening design basis, four platform categories are relevant, distinguished by payload capacity, warhead effect, and operational characteristics.

Class I: FPV and lightweight precision platforms. Warhead mass of one to five kilograms, typically shaped-charge configuration. Precision guidance to individual components. Primary threat to instrumentation, cable trays, valve operators, junction boxes, and other safety-significant items not protected by structural hardening. Low unit cost enables high sortie rates. Operator control-link dependent; AI-enabled variants are increasingly emerging.

Class II: Loitering munition attack drones. Warhead mass of twenty to seventy kilograms, general-purpose blast-fragmentation or shaped-charge configuration. Effective against electrical switchgear, transformer equipment, generator installations, above-grade piping, and external system components. Shahed-136 class platform represents the operationally demonstrated baseline. One-way expendable; can be deployed in coordinated waves.

Class III: Conventional aerial munitions delivered by uncrewed platforms. Warhead mass of one hundred to five hundred kilograms. Capable of causing structural damage to hardened installations. Less frequently observed in infrastructure campaigns but within the threat envelope of state-level adversaries employing armed UAS with conventional munition payloads.

Class IV: Autonomous AI-enabled systems. Characterized primarily by reduced dependence on operator control-link rather than by warhead class, AI-enabled variants exist across warhead classes I through III. Relevance to hardening: jamming-resistant autonomous platforms require physical and structural countermeasures because electronic disruption of the control link is ineffective. AI-enabled reconnaissance platforms with persistent loiter capability present a distinct surveillance threat independent of their strike function.

4.2 Infrastructure Vulnerability Profile

UAS threats exploit the above-grade exposure of safety-significant systems, structures, and components. The sector implementation annex for each sector defines the specific systems that constitute the safety-significant target set for that sector. Across all sectors, the common UAS vulnerability pattern is the exposure of above-grade support infrastructure on which the facility's protected operational state depends: external power connections and yard switchgear; backup generation equipment and external fuel supply installations; process cooling and heat rejection equipment; control and communications infrastructure routed through cable trays or above-grade conduit; and chemical or consumable storage critical to sustained operations.

The cumulative degradation scenario — in which repeated Class I strikes on individual above-grade components progressively reduce the facility's functional capacity — is the more realistic threat model for most infrastructure targets, as compared to a single decisive strike on a major hardened structure. The sector implementation annex shall define the consequence chain from cumulative UAS damage to the sector-specific loss-of-function threshold, including the intermediate states in which partial function is maintained. Still, the facility's ability to reach and sustain its protected operational state is progressively compromised.

4.3 Detection and Characterization

A facility implementing the DBA trigger framework must maintain a UAS detection and characterization capability as a standing element of its monitoring posture. Detection capability is a prerequisite for UAS-related trigger observables and for the protective actions specified in this document. A multi-sensor approach provides the most robust detection performance across UAS platform types and operational conditions.

Radar detection provides coverage against larger platforms at extended range and functions in adverse weather and low-light conditions. Limitations include difficulty in detecting small, low-radar-cross-section platforms, such as Class I FPV drones, at operationally useful ranges.

Radio-frequency (RF) detection identifies control-link emissions associated with remotely piloted platforms. Effective against communications-link-dependent systems; progressively less effective as AI autonomy reduces control-link use. RF detection provides attribution information — the presence of a control signal locates not only the platform but potentially the operator.

Acoustic detection identifies the characteristic acoustic signatures of rotary-wing UAS and some fixed-wing platforms. Effective at short range; performance degrades in high-ambient-noise environments.

Electro-optical and infrared (EO/IR) sensors provide visual and thermal detection and characterization. AI-enabled visual recognition systems — including smartphone-based applications capable of distinguishing drone platforms from other airborne objects in both day and night conditions using electro-optical infrared sensors — have been developed and deployed for military airbase protection. When integrated with ground-based LIDAR, such systems provide all-weather, three-dimensional, real-time UAS tracking through conditions that degrade conventional electro-optical sensors. These low-cost, rapidly deployable detection layers can supplement fixed multi-sensor arrays and are particularly relevant for extending detection coverage during the standing preparedness posture required at the initial tier of the sector trigger framework.

The facility's detection capability shall be specified in the site-specific UxS vulnerability assessment required under Section 9, with sufficient layering across modalities to provide confidence in detection against the threat classes present in the facility's threat environment. No single detection modality provides adequate coverage across the full UAS platform range.

5. Uncrewed Surface Vessels (USV)

5.1 Threat Characterization

The USV threat to critical infrastructure is primarily relevant to facilities sited on navigable rivers, lakes, reservoirs, or coastal waters. The same hydrological feature that provides process water, cooling water, or logistical access to a facility may also provide an adversary with an unobstructed waterborne approach corridor.

Class I: Small explosive-payload surface craft. Payload mass of twenty to one hundred kilograms. Accessible to sub-state actors using commercially available watercraft modified for payload carriage and remote guidance. Primary threat to waterfront infrastructure, intake screening structures, and above-waterline components of process water and cooling water systems.

Class II: Large explosive-payload attack USV. Payload mass of one hundred to five hundred kilograms. Demonstrated at the state and near-state level by Ukrainian Magura V5 class operations against Black Sea Fleet surface vessels and infrastructure. The warhead effect at this payload scale can cause severe structural damage to reinforced-concrete intake structures, pump-house foundations, and waterfront retaining walls. The structural damage mechanism differs from that of UAS strikes: rather than precision delivery to a specific component, a large USV detonation at or near an intake structure produces a combination of blast overpressure, hydrodynamic shock, and fragmentation loading on submerged and waterline-level infrastructure.

Class III: Surveillance and reconnaissance USV. The payload is a sensor suite rather than an explosive charge. The threat is to the facility's physical security perimeter and operational security rather than to the physical infrastructure directly. Reconnaissance USV can identify intake structure configurations, waterfront sensor positions, and patrol patterns in support of subsequent strike operations by USV or UUV platforms.

5.2 Infrastructure Vulnerability Profile

The primary critical infrastructure vulnerability to USV attack is the waterfront interface: intake structures, pump houses, screening facilities, and any waterline-level structural elements on which the facility's core processes depend. Facilities dependent on continuous waterway-sourced flow — whether for reactor cooling, water treatment intake, industrial process water, or offshore terminal operations — share the structural characteristic that critical infrastructure components are concentrated at or near the waterline and accessible from the waterway.

A secondary vulnerability common across sectors is the waterfront infrastructure that supports site access and logistics. Facilities that receive fuel, consumables, or emergency equipment by waterborne transport are dependent on waterfront loading and berthing infrastructure. USV damage to this infrastructure during a sustained threat period can interdict the resupply of consumables necessary to maintain the facility's protected operational state for its design basis duration.

The tertiary vulnerability, applicable to facilities with waterway-discharged effluent, is the discharge structure. Damage to the discharge path can impair heat-rejection or process-effluent management capabilities, regardless of the intake system's status. The sector implementation annex shall define the specific systems constituting the protected target set for each sector.

5.3 Detection and Characterization

USV detection must cover the waterborne approach corridor from a distance sufficient to enable protective response. The response time available against a USV attack is a function of the USV's approach speed and the detection range achievable by the facility's monitoring systems. At a representative approach speed of twenty-five to forty kilometers per hour and a detection range of two kilometers, the available response window is three to five minutes — sufficient for alarm activation and barrier system engagement but not for active interdiction by external response forces.

Surface radar provides detection of USV platforms at ranges consistent with the waterborne approach corridor. Performance against small, low-freeboard platforms in cluttered waterway environments is the principal technical challenge. Coastal surveillance radars optimized for small-target detection in maritime environments provide materially better performance than general-purpose perimeter radars for this application.

Electro-optical and infrared surveillance of the waterborne approach corridor supplements radar detection. Thermal signatures from USV propulsion systems are detectable at operationally relevant ranges. AI-enabled object recognition applied to the waterway approach has demonstrated the ability to distinguish USV platforms from recreational and commercial watercraft based on silhouette, speed profile, and approach trajectory.

Acoustic hydrophone arrays positioned across the waterway approach corridor can detect the acoustic signature of USV propulsion systems at ranges exceeding those achievable by surface radar for small-platform targets in some acoustic environments. Hydrophone detection is passive and does not reveal the presence of the surveillance capability to the adversary.

6. Uncrewed Undersea Vehicles (UUV)

6.1 Threat Characterization

The UUV threat to critical infrastructure is directionally significant and operationally nascent. No publicly documented case of a UUV attack against civilian critical infrastructure has occurred as of the date of this document. The threat is nonetheless included in this document for three reasons: the engineering consequence of a UUV attack on water intake infrastructure or submerged structural elements would be safety-significant and is not addressed by surface-level physical protection measures; the UUV capability base at state and increasingly near-state level is documented in open-source intelligence literature; and the design basis for a DBA addendum must address the threat trajectory, not only the demonstrated baseline, given the methodology's requirement to bound the design against future capability evolution.

Class I: Autonomous explosive-payload UUV. Payload mass of ten to one hundred kilograms. Capable of approaching a submerged intake structure along the bottom of the waterway and detonating at contact range or at a pre-programmed standoff distance. The hydrodynamic shock from an underwater detonation against a reinforced-concrete or steel intake structure is qualitatively different from the above-water blast effects that surface-hardening measures address. Different engineering parameters govern underwater detonation effects and require separate structural analysis.

Class II: Mine-emplacement and persistent-threat UUV. Payload is a mine or improvised explosive device positioned on or adjacent to submerged infrastructure. Unlike an attack UUV that delivers its payload in a single event, a mine-emplacement UUV creates a persistent threat that activates at a time or under conditions determined by the adversary. Detection of emplaced mines requires active subsurface surveillance of all submerged infrastructure, not merely detection of the delivery platform as it approaches.

Class III: Survey and reconnaissance UUV. The payload is a sensor suite for mapping submerged infrastructure—threat to operational security and targeting intelligence, not to physical infrastructure directly.

6.2 Infrastructure Vulnerability Profile

The primary vulnerability common across sectors is any below-waterline infrastructure on which the facility's protected operational state depends: submerged intake piping and screening structures; below-waterline foundation elements of waterfront structures; submerged sections of process piping systems; and electrical or communications cable runs that cross waterways in submerged configurations. A structural breach below the waterline initiates consequence chains that are more complex to manage than above-grade damage because the damage location is inaccessible for inspection and repair during an active threat period.

The sector implementation annex shall define the specific submerged systems that constitute the protected target set for each sector and specify the consequence chain from UUV-induced structural breach to the sector-specific loss-of-function threshold.

6.3 Detection and Characterization

UUV detection is technically demanding and represents a capability gap for most facility physical protection programs, which were designed around surface-level threats. The minimum detection capability appropriate to the DBA threat environment consists of passive acoustic hydrophone arrays positioned to monitor the submerged approach corridor, supplemented by periodic active sonar sweeps of intake structures and their immediate approach zones. Continuous subsurface video surveillance of critical intake structure entrances provides the most direct detection of both approaching platforms and emplaced devices, but requires clear water visibility and maintenance access.

Given the nascent state of the UUV threat and the immaturity of civilian facility underwater surveillance practice, the site-specific UxS vulnerability assessment required under Section 9 shall address UUV detection as a gap assessment item: identifying the current subsurface surveillance posture of the facility, the detection capability achievable against the relevant UUV threat classes given the facility's hydrological and physical environment, and the measures required to establish a credible detection capability consistent with the facility's threat environment.

7. Cross-Domain Compounding

The treatment of UAS, USV, and UUV as separate analytical sections should not obscure the operationally significant scenario in which platforms from two or more domains are employed in coordinated attack packages against a single facility. Cross-domain compounding produces consequence chains that are more severe than those from single-domain attacks and challenge the engineering design assumptions underlying single-domain hardening measures.

7.1 UAS and USV Coordinated Attack

The most operationally accessible cross-domain scenario combines UAS and USV in a simultaneous or sequentially staged attack. The Ukraine conflict has demonstrated coordinated operations in which UAS provide reconnaissance and distraction — engaging air defense sensors and response assets — while USV platforms approach the primary target on the surface. Applied to a critical infrastructure facility, this scenario presents the following compound challenge: the UAS component engages above-grade protective systems, fire suppression, and response personnel; the USV component exploits the resulting degraded defensive posture to approach the waterfront. The consequence chain involves the simultaneous degradation of aerial and waterborne protective measures, damage to above-grade infrastructure from UAS strikes, and compromise of the waterfront system from USV detonation.

The engineering design response to the UAS-USV compound scenario requires that the hardening of above-grade systems and the physical protection of the waterborne approach corridor be designed and evaluated together, not independently. A hardening design that adequately addresses each domain in isolation but creates an interdependency — where the waterfront passive barrier system requires above-grade power or communications to activate, and the UAS component disables that power first — fails the local sovereignty of safety functions principle and must be redesigned.

7.2 UAS and UUV Sequential Attack

A more technically demanding scenario involves UAS strikes on above-grade systems followed by UUV exploitation of the resulting degraded monitoring posture. If UAS strikes on the facility's above-grade surveillance infrastructure also disable the surface-level components of the facility's waterborne detection system, the subsequent UUV approach may occur against a degraded detection baseline. This scenario reinforces the requirement that subsurface detection capability — passive acoustic arrays and subsurface camera systems — must be powered from protected, dedicated circuits that are not co-located with the above-grade surveillance systems that UAS strikes would target.

7.3 Cyber-Kinetic Integration

Within the UxS context, the cyber dimension is relevant to detection and response. UxS platforms operating with AI autonomy may be coordinated with a cyber intrusion that degrades the facility's UxS detection systems immediately before the physical attack. A cyber event that silences radar, disables RF detection, or corrupts the sensor fusion system feeding the facility's monitoring posture creates a detection gap that the UxS physical attack then exploits. The sector trigger framework observables must include cyber-physical system anomalies as indicators that a UxS attack may be imminent, and detection systems must maintain adequate redundancy and independence to preserve detection capability through cyber-induced degradation of individual sensor modalities. The Three Threads of Cyber Security companion document addresses the cyber-kinetic combined-attack threat category for the nuclear sector; analogous cybersecurity instruments should be developed for each sector's implementation.

7.4 Sector-Specific Compounding Scenarios

The sector implementation annex for each sector shall develop the cross-domain compounding scenarios that are most operationally relevant to that sector's facility type and threat environment. The scenarios in Sections 7.1 through 7.3 are sector-agnostic frameworks; their specific application — including identifying the above-grade systems that UAS would target, the waterfront systems that USV would target, the submerged systems that UUV would target, and the interdependencies among them — is a sector-specific analysis task.

8. Hardening Design Principles for UxS Threats

The hardening design principles established in the applicable sector framework documents define the general engineering framework for military action resilience. This section develops those principles specifically for the UxS threat environment across all three domains, providing design basis guidance sufficient to inform facility-specific vulnerability assessments and hardening designs. These principles are expressed at the level of engineering design intent; specific design criteria, structural performance parameters, and acceptance standards will be developed in classified supplements and in the facility-specific analyses required under the applicable sector implementation annex.

These principles are to be applied as an integrated design framework, not as independent requirements to be satisfied in isolation. The adequacy of any individual hardening measure shall be assessed in the context of the full cross-domain threat profile and in recognition of the compounding scenarios described in Section 7. A hardening design that satisfies each principle in isolation but fails under the compound scenario has not met the intent of this section.

Note on design basis currency: these hardening design principles are not static. The UxS threat evolves on a timescale that can outpace the periodic safety review. The mandatory review cycle established in Section 9 governs the intervals at which the threat-class assumptions underlying these principles must be re-evaluated and the hardening design basis updated accordingly. The principles below are written to be durable across expected near-term UxS capability evolution; the threat-class parameters used to apply them must be refreshed at the intervals and upon the triggers specified in Section 9.

8.1 Physical Hardening and Below-Grade Installation

Systems, structures, and components (SSCs) credited in the sector-defined protected operational state analysis shall be protected against the effects of the warhead classes characteristic of the UxS threat environment at the facility. The applicable threat class for each SSC shall be established in the site-specific UxS vulnerability assessment, taking into account the adversary profile, documented UxS capability inventory, and the SSC's function and location. The hardening design basis shall be set at the threat class representative of the highest-capability adversary with plausible access to the facility.

Below-grade installation is the preferred hardening approach for SSCs that can practicably be located underground. Burial or bunkering provides the highest degree of protection against all three UxS domains: it removes the component from direct aerial observation (defeating the UAS reconnaissance-strike cycle), places it below the waterline effect zone of surface detonations (reducing USV threat exposure), and moves it away from the submerged infrastructure zone where UUV threats concentrate. SSCs that cannot practicably be located below grade shall be addressed through structural hardening sufficient to maintain their credited function under the applicable threat class loading, with supplementary protection from passive kinetic barriers where structural hardening alone is insufficient.

The blast and fragmentation resistance of above-grade hardened enclosures shall be evaluated against the cumulative loading of repeated UAS strikes at the same location, not only against a single-event loading. The UxS persistent and cumulative threat characteristic (Section 3.3) requires that structural evaluations address the degradation of protective capacity under repeated loading cycles. A structure that survives the first strike may not survive the fifth if its protective capacity has been progressively reduced.

For USV threats, hardening of waterfront infrastructure shall address underwater blast effects as well as above-water blast and fragmentation. The structural performance requirements for below-waterline elements shall be derived from the design-basis USV detonation scenario at a standoff representative of the minimum distance achievable against the facility’s waterfront protective measures. The hardening and barrier designs are therefore interdependent and shall be developed together.

For UUV threats, the structural performance of submerged infrastructure shall account for underwater detonation effects at contact or near-contact ranges. The engineering analysis shall apply hydrodynamic shock loading appropriate to the UUV Class I payload mass and detonation geometry.

8.2 Passive Kinetic Standoff Barriers

Passive kinetic standoff barriers are physical installations that enforce a minimum detonation distance between an inbound UxS platform and a protected SSC, without dependence on external power, communications, or operator action to perform their protective function. This passive character is the defining design requirement: a barrier that requires activation, a power supply, or a command signal to deploy is not a passive kinetic barrier for purposes of this section and does not satisfy the local sovereignty-of-safety-functions principle.

Passive kinetic barriers are sacrificial by design. A barrier performs its function by inducing early detonation of the incoming warhead at the barrier structure rather than at the protected SSC. The consequence is that a strike event that engages a barrier section destroys that section.

Until the affected section is restored from the on-site replacement materials inventory, the protected SSC reverts to unshielded exposure. The design and operational requirements for passive kinetic barriers, therefore, have two inseparable components: the initial installation design and the post-strike inspection and restoration protocol.

For UAS threats, passive kinetic barriers typically take the form of mesh and cable arrays positioned above or around protected SSCs to intercept incoming platforms before they reach the protected component. The mesh parameters, standoff distance, and structural configuration shall be derived from the facility's design-basis UAS threat class.

For USV threats, passive kinetic barriers take the form of physical boom systems, anti-approach obstacles, and reinforced buoy chains positioned across the waterborne approach corridor at a standoff distance from critical waterfront infrastructure sufficient to attenuate design-basis USV detonation effects below the structure's damage threshold. Barriers shall be anchored to fixed points and shall be visible and lit for navigation safety.

For UUV threats, passive kinetic barriers take the form of physical net barriers and grating structures positioned across intake channels or waterway approaches at depths sufficient to intercept bottom-following and mid-water UUV platforms.

The on-site replacement materials inventory for all passive kinetic barrier installations shall be maintained as a required consumable. Inventory shall be sufficient to restore all design-basis barrier installations following a strike event that destroys one or more barrier sections, and shall be assessed at each monitoring cycle.

8.3 Separation, Dispersal, and Redundancy

Redundant trains of safety-significant systems shall be physically separated and, where practicable, located in different structures, to prevent common-cause failure from a single UxS strike. The separation distance shall be sufficient that a single Class II UAS or USV detonation event does not simultaneously affect two redundant trains of the same safety function. The site-specific vulnerability assessment shall map the current physical configuration of redundant safety system trains against the blast radius of the threat class.

Consumable storage — fuel, process chemicals, barrier replacement materials, and other sustainment inventories — shall be dispersed across protected storage locations rather than concentrated in a single facility. The dispersal design shall ensure that a single UAS strike targeting the primary consumable storage location does not reduce the total on-site inventory to a level below the quantity required to sustain the protected operational state for its design-basis duration. The dispersal requirement applies specifically to the UAS reconnaissance-strike dynamic: persistent aerial surveillance can identify concentrated storage as a high-value target.

8.4 Concealment and Signature Reduction

Safety-significant systems that cannot be practicably hardened or located below grade shall, where feasible, be concealed from aerial, surface, and subsurface observation through the use of natural or constructed visual barriers, below-grade installation of externally observable components, and camouflage of above-grade installations against both visual and thermal detection.

Thermal and electromagnetic signatures of operating equipment shall be assessed for detectability by the sensor modalities carried by the adversary's reconnaissance platforms. For UAS reconnaissance, relevant signatures include thermal emissions from generation and cooling equipment, electromagnetic emissions from switchgear and control systems, and the visual footprint of above-grade structures. For USV reconnaissance, relevant signatures include thermal emissions visible from surface surveillance sensors and visual identification of waterfront infrastructure from the water's surface. For UUV reconnaissance, relevant signatures include the acoustic signatures of pump motors and rotating machinery transmitted through the waterway and detectable by a passive acoustic survey platform.

Concealment does not replace hardening; it complements it by increasing the number of reconnaissance sorties the adversary must expend to identify and characterize the target and by raising the probability that replacement equipment installed during the protected operational state period can be positioned without immediate targeting.

8.5 UxS Detection as a Hardening Prerequisite

Detection capability is a prerequisite for hardening effectiveness, not an alternative to it. A facility that has installed passive kinetic barriers, below-grade storage, and hardened redundant trains but cannot detect an approaching UAS, USV, or UUV platform in time to initiate protective procedures has not satisfied the engineering requirements of the DBA framework. Detection is the initiating condition for the protective response; hardening is the engineering basis for the consequence limit if detection fails or if the attack proceeds faster than the available response window.

The multi-sensor detection requirements for each domain are addressed in Sections 4.3, 5.3, and 6.3. For hardening design purposes, the key parameter is the detection range-to-response-time relationship: the detection system must provide adequate warning time to initiate the protective procedures specified in the facility's operating procedures for its protected operational state. Where the detection range achievable against the design-basis platform class does not provide sufficient warning time for a given protective procedure, the hardening design must provide protection even in the absence of warning.

8.6 Hardening Assessment Integrated with the Review Cycle

The hardening design basis is not a static engineering determination made at initial implementation and thereafter fixed. The UxS threat evolves on timescales measured in months to years, driven by technological development, battlefield experience, and the competitive dynamics of military innovation. The mandatory review cycle established in Section 9 — two-year base interval for high-threat-profile facilities, three-year interval for standard facilities, and four explicit intelligence triggers for unscheduled review — governs the intervals at which the threat-class assumptions underlying the hardening design basis must be re-evaluated.

Each scheduled review and each intelligence-triggered review shall include a hardening adequacy assessment addressing the following questions. First, have adversary UxS capabilities evolved in ways that move the design-basis threat class above the class used in the current hardening design? If so, the hardening design requires an update. Second, have passive kinetic barrier installations been subjected to strike events that consumed barrier sections and not yet been fully restored? If so, restoration shall be completed before the review is closed. Third, has the dispersal and concealment posture been compromised by operational changes that have made previously concealed or dispersed assets visible or consolidated? If so, the operational security posture shall be restored. Fourth, has the detection capability been verified against the current threat class within the review cycle?

9. Review Cycle Requirements

The UxS threat environment evolves on a timescale that is inconsistent with the decadal safety review cycle used for most external hazard categories. The mandatory review cycle established in this section applies to all sector implementations of the DBA-UxS framework.

9.1 Mandatory Review Intervals

Facilities assessed as operating in a high-threat-profile environment — defined as facilities for which a plausible near-peer adversary has documented UxS capabilities and assessed motivation to use them against the facility or its category — shall conduct a full UxS threat reassessment at intervals not exceeding two years. Facilities in standard-threat-profile environments shall conduct the reassessment at intervals not exceeding three years. Facilities shall not self-assess their threat profile; the competent national regulatory authority shall determine the applicable classification in coordination with national intelligence and defense establishments.

The UxS threat reassessment shall address all three domains — UAS, USV, and UUV — even when the primary assessed threat to the specific facility is concentrated in a single domain. Capability developments in any domain may be relevant to the cross-domain compounding scenarios described in Section 7, and capability developments in one adversary's inventory may be relevant to other adversaries' future capability trajectory. The reassessment shall also address the hardening adequacy questions specified in Section 8.6.

9.2 Intelligence Triggers for Unscheduled Review

Regardless of the scheduled interval, any of the following four intelligence triggers shall initiate an unscheduled UxS threat reassessment within ninety days of the triggering event. The facility's competent regulatory authority shall be notified of the trigger and the initiation of the unscheduled review within thirty days.

Trigger A: Confirmed acquisition or deployment by a relevant adversary of a new UxS platform class, variant, or capability — including AI-enabled autonomous operation, improved warhead mass, improved range, or improved countermeasure resistance — that moves the threat class above the design-basis class used in the facility's current hardening assessment.

Trigger B: Demonstrated operational employment of UxS against energy infrastructure, water system infrastructure, data center infrastructure, or comparable critical infrastructure targets by a relevant adversary in any theater of operations, indicating doctrine, intent, and operational capability that may apply to the facility category.

Trigger C: Credible specific threat reporting — from classified or unclassified intelligence sources — indicating an adversary's intention to conduct UxS operations against the facility, its facility category, or infrastructure in the host nation or allied nations.

Trigger D: Material deterioration in the geopolitical environment — including significant deterioration of bilateral relations with a UxS-capable state, mobilization or forward deployment of UxS-capable adversary forces, or declaration of a threat condition affecting the facility's threat environment by the host nation's national defense establishment — that materially changes the assessed probability that the design-basis UxS threat will be realized.

9.3 Classified Assessment Integration

The unclassified analysis required by this section shall be supplemented by a classified UxS capability assessment from national intelligence authorities at each scheduled review cycle. The facility shall request this assessment through the monitoring and data-sharing pathways established under the applicable sector Level 4 Monitoring Guidance. Where a facility's own analysis of open-source reporting identifies a potential Trigger A or Trigger B condition, the facility shall request a classified confirmation assessment through those pathways before concluding that the trigger has been met.

10. Sector Implementation Annex Architecture

This document establishes the cross-sector foundational content of the DBA-UxS framework. Sector-specific content — including the definition of safety-significant systems, facility-specific vulnerability profiles, sector-specific consequence chains, the definition of the sector-protected operational state, hardening requirements tied to sector-specific targets, and parent document cross-references — is addressed in sector implementation annexes that inherit from this document.

Each sector implementation annex is a child document of this instrument. The annex shall not repeat content that is addressed in this document. The annex shall address, at a minimum, the following elements for its sector:

Safety-significant system definition: The specific systems, structures, and components that constitute the safety-significant target set for the sector. This is the sector equivalent of the nuclear sector's SSC concept as applied to safety functions.

Sector-protected operational state: The sector-specific condition that the facility must be capable of reaching and sustaining under UxS attack. This is the sector equivalent of the nuclear sector's Autonomous Safe Shutdown Condition.

Facility-specific vulnerability profiles: The application of the generic vulnerability profiles in Sections 4.2, 5.2, and 6.2 to the specific systems and facility types of the sector.

Sector-specific consequence chains: The consequence chain from UxS strike to sector-specific loss of function, including intermediate states.

Sector-specific hardening targets: The application of the hardening design principles in Section 8 to the specific safety-significant systems of the sector.

Parent document cross-references: The specific parent documents within the sector's document series to which this document's provisions relate.

Sector-specific regulatory and standards references: The regulatory framework, technical standards, and sector-specific guidance documents applicable in the sector implementation.

The following sector implementation annexes have been developed or are in development:

DBA-MA-UxS-NA v1.0 (Nuclear Implementation Annex) — Released concurrently with this document. Covers nuclear power plants and supporting nuclear infrastructure.

DBA-W-UxS (Water Sector Implementation Annex) — In development.

DBA-DC-UxS (Data Center Implementation Annex) — In development.

DBA-OIL-UxS / DBA-GAS-UxS (Petroleum Sector and Natural Gas Sector Implementation Annexes) — Planned.

11. References

11.1 Defense and Technical References

NATO Joint Air Power Competence Center, Uncrewed Aircraft Systems: Terminology, Definitions, and Classification, JAPCC, Kalkar (current edition).

U.S. Department of Defense, Uncrewed Systems Integrated Roadmap, Office of the Under Secretary of Defense for Research and Engineering (current edition).

Open-source technical literature on Shahed-136 performance characteristics, Ukrainian naval drone operations (Magura V5 and successor platforms), and AI-enabled autonomous UAS operations as referenced in classified supplements to this document.

11.2 Sector Implementation Annexes

DBA-MA-UxS-NA (v1.0) — Nuclear Implementation Annex: Uncrewed Systems Threat Considerations for Nuclear Facilities.

DBA-W-UxS — Water Sector Implementation Annex: Uncrewed Systems Threat Considerations for Water Infrastructure (in development).

DBA-DC-UxS — Data Center Implementation Annex: Uncrewed Systems Threat Considerations for Hyperscale Data Center Infrastructure (in development).